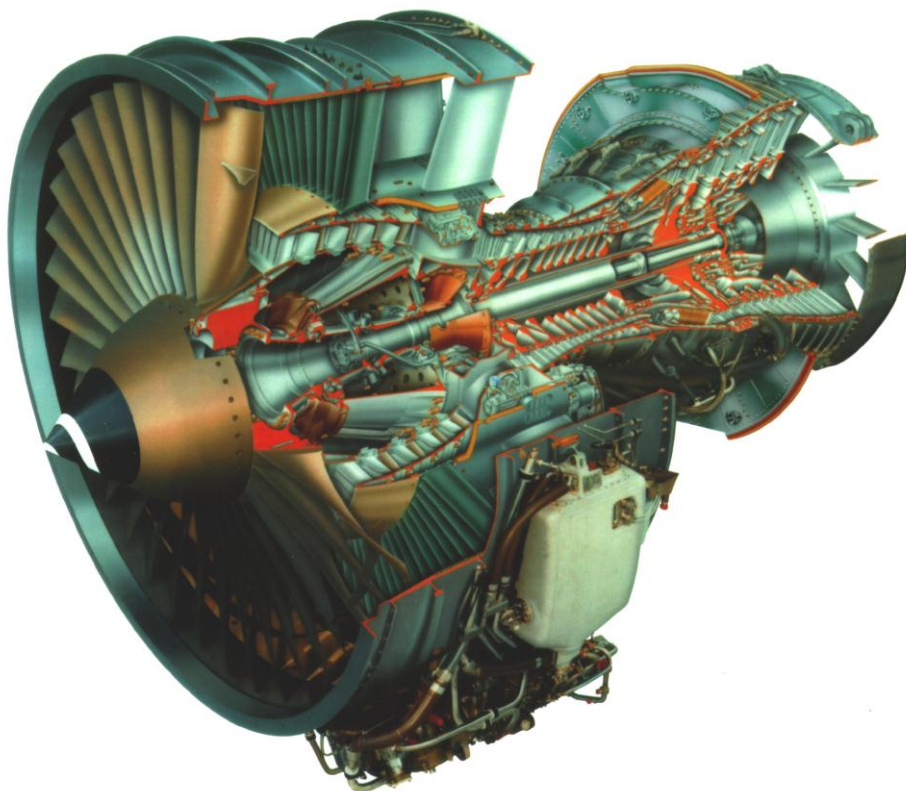


Phase B: Feasibility Study of CFM56 Engine Family Test Cell



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1. INTRODUCTION

Regarding test cell facilities with wide range of thrust measurement, there is a couple of test cell facilities with capability of testing several types of engines families, but it should be considered that their components have been designed and manufactured based on comprehensive studies and deliberation in several aspects of location and environmental considerations, supply systems capabilities, constructions, engine belongings and hardware, auxiliary systems, software and so on. As well as, some systems are specified for only one type of engine and extra ones must be provided for any types of engine. All these items make these procedure contingent on spending massive engineering, cost, time, manpower, infrastructure and etc. So it is not possible to modify and develop a certain design for specific engine family to another one with higher thrust range with simple modification and almost the same cost, therefor it is possible to design and manufacture class 100 k lbf test cell in order to test CFM56 and CF6 families but feasibility studies shown that lots of reworking required in majority of parts which is described in detail in this report.

The following are clarifications of the terms used within this report. They reflect the definitions generally accepted in the industry.



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2. DEFINITIONS

Intake

The intake provides a uniform air flow to the engine inlet and absorbs part of the noise generated. The inlet may typically pass through a bird screen, noise reduction splitters and a flow conditioning screen before entering the cell testing section.

Turning vanes

Turning vanes installed to the inlet and exhaust sections of the test cell can reduce the head loss and uniformly guide the fluid flow. Turning vanes are designed to assist in redirecting the inlet air flow from a vertical flow to a horizontal flow in a uniform manner.

Airflow Control Screen

The flow screen is installed across the entire cross section of the test cell downstream of the intake stack. The open area of the flow screen is designed to provide the optimum pressure drop conditions in a test cell.

Engine Inlet

Once the air in the chamber room is accelerated into the engine, it passes through a section called inlet which provides a uniform air flow to the engine and absorbs part of the noise generated.

Test chamber

Test chamber is an enclosed space prepared and well equipped for engine test plans and such activities as, measuring components performance quantities, altitude condition simulation, and engine monitoring.

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Augmenter Tube

Augmenter tubes serve two purposes: to cool the exhaust gases by mixing them with fresh air, and to attenuate noise.

Blast basket

It enhances audible noise reduction by shifting the exhaust noise low frequencies to higher and better managed frequencies.

Exhaust

The exhaust section is an aligned section to direct jet exhaust of the engine at least acoustic emission. It consists more often of an augmenter, a diffuser section followed by a boot section and an exhaust stack

Intake and exhaust silencer

Intake and exhaust silencer are designed to meet the noise requirements without imparting excessive cell depression and exhaust backpressure.

Stack Sound and Treatment Systems

The exhaust stack allows the diluted combustion streams to be expelled through the atmosphere with aerodynamic noise pollution. The designs of the intake and exhaust acoustic treatments should provide a uniform air flow at the chamber room, which is where the engine performance measurements are taken.

Superstructure

The superstructure will interface between the test cell structure and the top of the thrust frame. It consists of four welded assemblies bolted to the beams embedded in the test cell ceiling. The superstructure provides the correct vertical positioning of the thrust stand rigidly bolted to the superstructure.

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Thrust Stand

Thrust Stand also known as framework, thrust frame, cradle or even bed, is a structure that holds an adapter mechanism connected to the engine.

Flexure plate

Flexure plates act as flexure hinges in order to transfer axial thrust to a load cell, but must support rigidly any other degrees of freedom.

Adapter

An adapter mechanism makes the interface between the engine and the floating part of thrust stand; the adapter is specific to each engine model and its instrumentation also allows to obtain engine data needed to acknowledge its performance status, analyzed from the test cell control room.

Bellmouth

It is a well-shaped contour used for an air flow smooth intake.

Test Nacelle

Nacelle is the engine outside envelope which defines the configuration of the secondary flow streamline.

Work Platform

It is a safe, stable work surface for the operators regardless of elevation height.

Lifting platform

It is a platform used in the test cell to lift engine, equipment's and tools along with technicians/engineers to a working height.

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Load cell

A Load Cell is a device that is used to measure weight or force in a number of different applications. The load cell itself is a transducer that is used to convert a force into an electrical signal.

Correlation Factor

A multiplier or an algebraic difference that is normally used to adjust corrected performance parameters measured at the facility to be correlated. This adjustment accounts for the difference in engine performance between the facility to be correlated and the reference facility.

Cell by pass ratio

The ratio of flow bypassing the engine relative to the flow that enters the engine is called the cell bypass ratio.

Cell aspect ratio

Cell aspect ratio is the ratio of the cross section area of the cell to the intake area of the engine.

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3. CIVIL CONSTRUCTION

Test cells are designed to recreate within its enclosure the flow conditions of steady air. As shown in Figure 3-1 any test cell comprises three main sections: the intake, the test main chamber and the exhaust region. The intake provides a uniform air flow to the engine and absorbs part of the noise generated; the smooth air flow is important to cool the room's ambience before it enters and is pumped through the augmenter. The designs of the intake and exhaust acoustic treatments should provide a uniform air flow at the chamber room, where the engine performance measurements are taken. The construction should also be aligned with the prevailing wind direction to prevent re ingestion of the engine combustion products emitted from the exhaust stack.

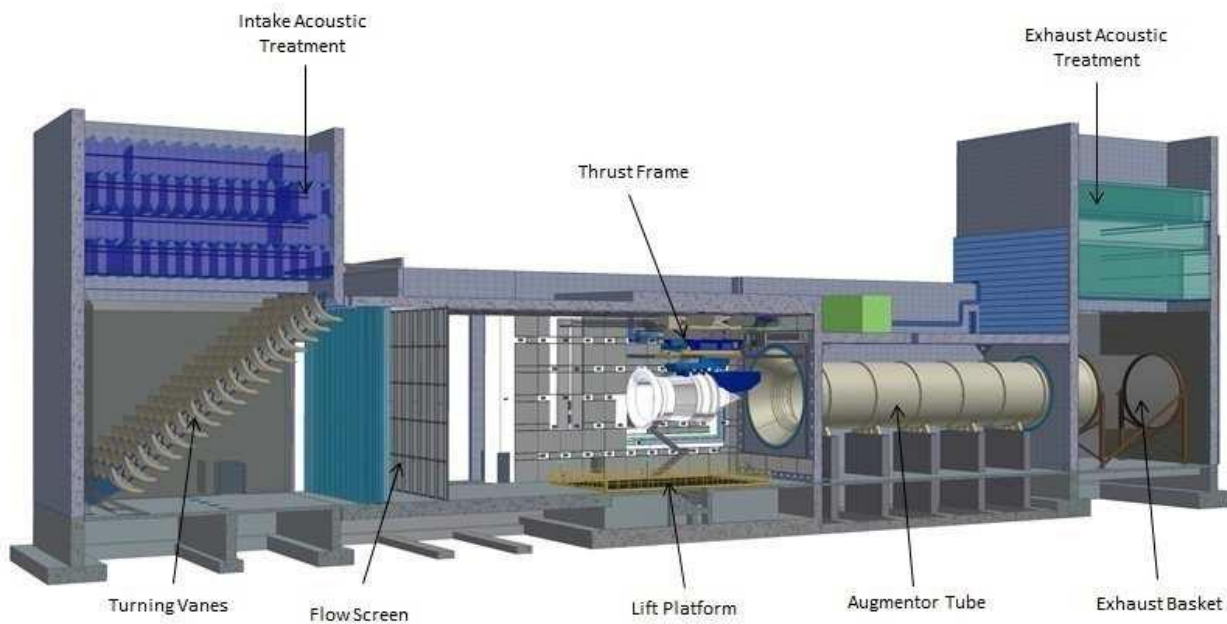


Figure 3-1: A typical test cell design

The test cell shown in Figure 3-1 has vertical inlet and outlet stacks. The advantages of this arrangement are: the noise radiation at ground level is lowered, clean air is sucked in with less danger of foreign object damage and erosion of the

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acoustic walls. The construction of a horizontal inlet channel and/or outlet stack is less expensive but it may be necessary to augment the volume of noise absorption materials in order to meet the required noise attenuation. Additionally, the air flow from a horizontal intake to the engine can be strongly influenced by external wind behavior. On the other hand the design of the vertical intake can take advantage of the 90° turn in the air flow to largely eliminate shear effects due to wind direction.

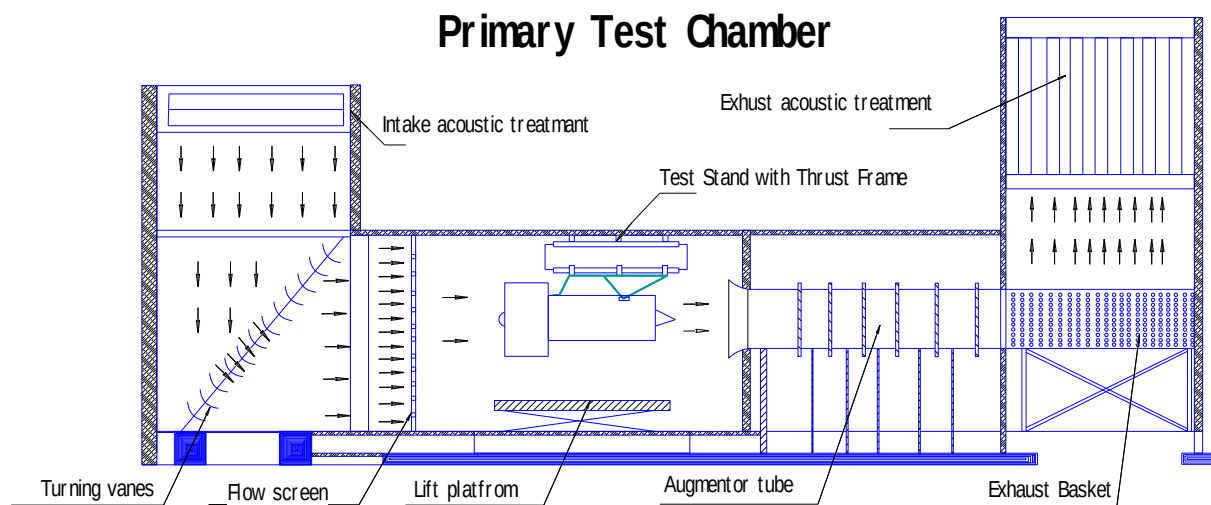


Figure 3-2: Schematic of test cell

3.1 TEST CELL

Test cells present some challenges e.g. flow separation, sink vortices ingestion, re-ingestion of the exhaust gases and resonance phenomena. These problems compromise the engine performance and have to be addressed when aiming to test more powerful engines.

Once the air in the chamber room is accelerated into the engine, wall-flow separation may occur if the cell remaining air flow is not high enough to prevent low flow velocities at the cell walls. Also, since air is drawn from all the directions nearby of the engine inlet, the components of the velocities may cancel beneath it: the flow angular

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velocity and the stagnation point originated are a perfect opportunity for vortices to form. This phenomenon depends on air and engine speed and its height above the floor. It is known that bigger engines lead to a lower peripheral air flow available thus increasing the risk of vortex formation. On the other hand, for a given test cell size large engines present problems related to

- Excessive noise and forces applied to the cell structures due to high air velocities;
- Thrust correction increase with increasing peripheral air flow.

Conclusion

The design of an indoor test cell has important consideration such as flow separation, sink vortices ingestion, re-ingestion of the exhaust gases and resonance phenomena. The components of the cell contribute to lower the cell pressure. To avoid significant flow distortion, this should be limited. It has important effect on the engine work conditions and the correction factor of the measurements. The design of an indoor test cell for different engine types should be performed considering the maximum mass flow and engine dimensions. In this way, more time and cost should be spent in design and construction steps of multi-engine test cell comparing with the test cell for one family of engines.

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4. TEST CELL COMPONENTS

In this section, the main components of each engine test cells and their design considerations are described. Also, the feasibility study of multi-engine test cell construction has been performed and compared with the test cell for one family engine testing in terms of design complexity, time and cost.

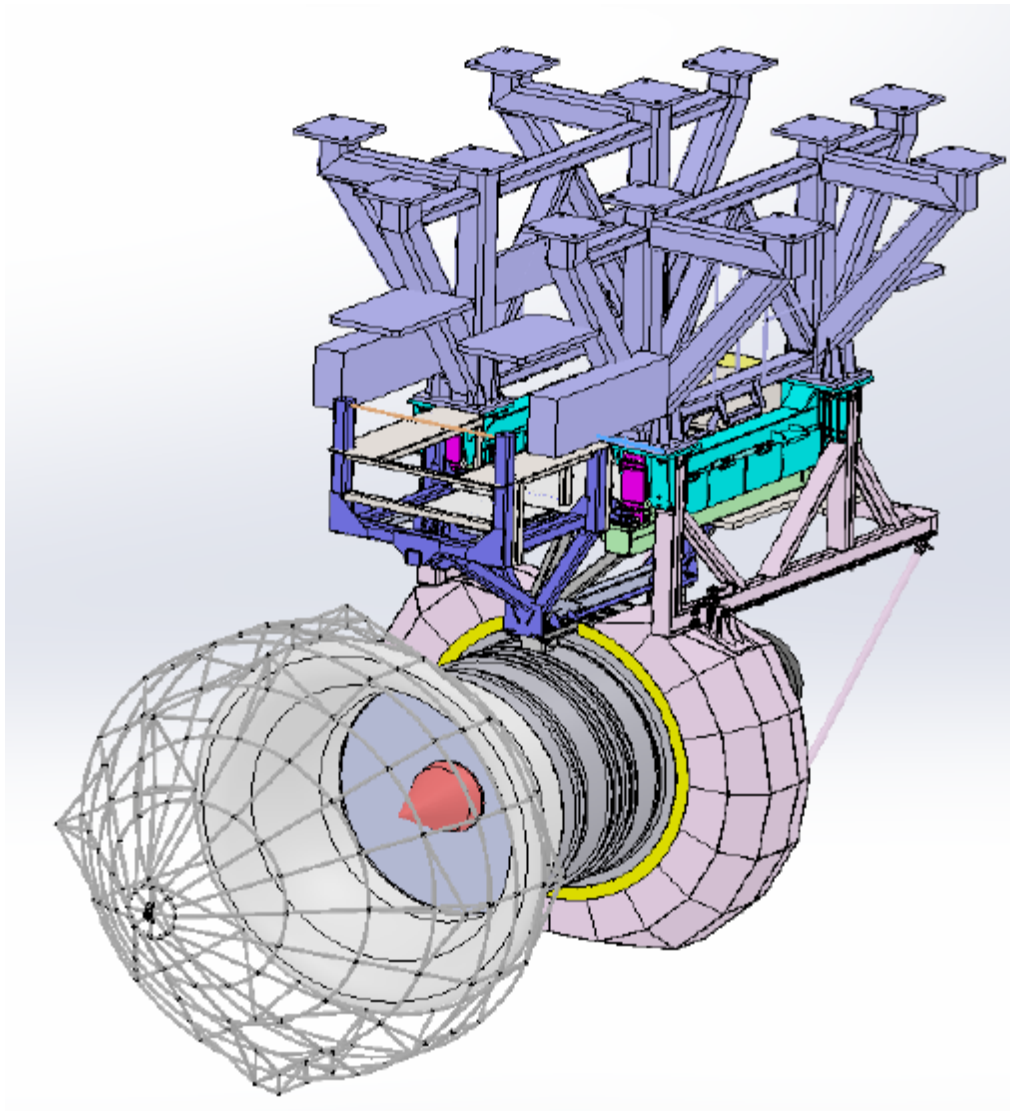


Figure 4-1: Overall view of thrust stand and engine

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4.1 ACOUSTIC SYSTEM

4.1.1 INLET ACOUSTIC SYSTEM

The silencer baffles are required to reduce engine noise emitted from the intake to acceptable levels add considerably to aerodynamic losses in the inlet region, because the restricted width of the channels necessary to produce sound attenuation introduces additional friction to the airflow.

The intake is acoustically lined in one to three rows of splitters. The needed noise attenuation quality is dependent on the thickness of the splitters and the distance between them. The larger distance and thickness are used to attenuate low frequency noise whereas high frequency noise is best attenuated by small distance and thickness.

Generally the dimensions of the silencer depend upon:

- The required total air mass flow
- The maximum cell depression
- The detailed design of the splitters.



Figure 4-2: Silencer

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4.1.2 EXHAUST ACOUSTIC SYSTEM

In design of the exhaust acoustic system, there are some important concerns in order to give the appropriate performance for different engine testing. These concerns which are listed below also influence on the design of other components of exhaust system such as augments and blast basket.

- Acoustic absorbents used in exhaust systems have an upper temperature limit of around 400°C. So it is necessary to introduce a significant degree of cooling either using water cooling or by having a large secondary air flow.
- In order to improve the efficiency of the anti-noise treatment and control the back pressure, the velocity has to be limited at 30-40 m/s along the exhaust stack.

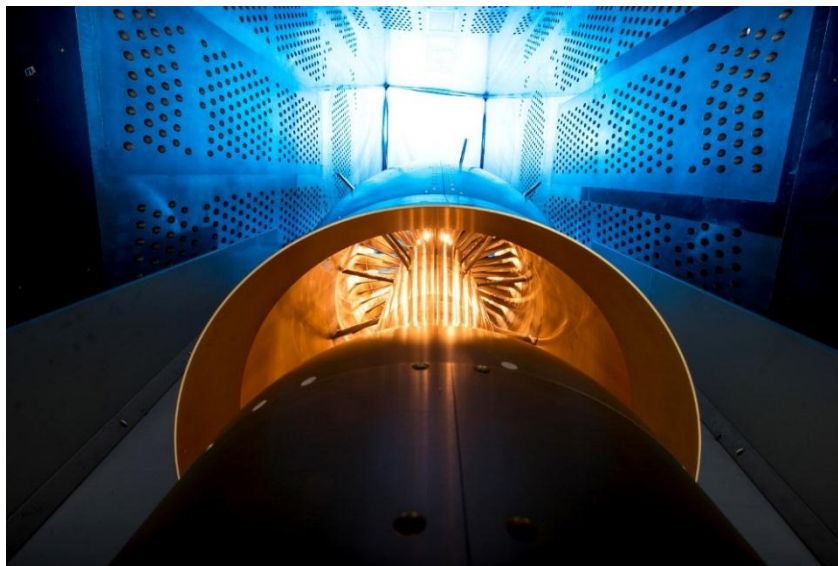


Figure 4-3: Exhaust sound and treatment system

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Conclusion

Generally, the acoustic absorbents of the test cells with the capability of testing the different types of engine should be designed considering the maximum range of mass flow and consequently high amount of noise. In this case, testing the engines with lower mass flow encounters the recalculation of pressure loss, thrust correction factors and influences on error level in measurements. In this way, more time and cost should be spent in design and construction steps of multi-engine test cell comparing with the test cell for one family of engines.

4.2 INTAKE SYSTEM

Typically, air intakes are vertical or horizontal. The balance is between a smoother and more uniform air flow to the engine with a horizontal intake because the flow is directly to the engine without having to turn through 90°, or fewer ground effects and less dust ingested with the vertical intake which can be several meters above ground level. Additionally, because the airflow from a horizontal intake to the engine can be strongly influenced by external wind behaviour it is usual to shield such intakes from cross winds by building side walls in front of the test chamber. Unfortunately, this can help generate eddy phenomena in the airflows close to the inlet; just the type of flow distortion the inlet should be designed to avoid.

On the other hand the design of the vertical intake can take advantage of the 90° turn in the air flow to largely eliminate shear effects due to wind direction.

Horizontal intakes are usually less likely than vertical intakes to re-ingest engine combustion products emitted from the exhaust. Not only must the physical separation of inlet and exhaust be adequate, it is also important to avoid placing the exhaust stack upwind of the inlet, particularly if the inlet is to be vertical.

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Within the constraints of the site, to avoid wind shear effects in the intake region the orientation of the test cell should be such that the intake is aligned with the prevailing wind direction. This will minimize but not eliminate the problem. The vertical intake has been chosen for the new test bed currently being used by the manufacturer because recent advances in the design of flow splitters and straighteners have substantially reduced the adverse effects.

Wind effects and pressure losses across the inlet section could affect the engine performance and possibly damage engine components. The drop in total pressure across the inlet system could result in lower pressure inside the cell. An excessive depression in cell pressure could affect the air flow stability around the engine and accuracy of measurements, and might possibly cause structural problems. The depression is due to the dynamic pressure loss resulting from the velocity in the test-bay and from the flow pressure losses in the air intake duct (flow open area, bends, acoustic panels, turning vane and screens).

It is generally recommended to limit the depression to a maximum of 150 mmH₂O. The reason is not the structural load limit but the correction on the measured parameters and the operation of the engine in conditions similar to those existing in open air test stand. At this limit the engine will be working in conditions close to those of free air, with minimum corrections necessary during measurements and their correlations. Non-turbulent intake air is crucial to proper, reliable aero engine testing. A proper inlet system must be designed in conjunction with the exhaust of the facility to achieve total cell flow with minimal cell depression. The distortion depends on the duct shape of the air inlet stack, the flow dividing obstacles and their profiles (acoustic panels), the shape and number of bends, the number and profile of the turning vanes, the wire

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mesh and bird duct screens, the flow length between the downstream edge of the panels or airflow straighteners and the engine inlet where vortexes and wakes are damped.



Figure 4-4: (a) Turning vanes and (b) Air flow control screen

Conclusion

To reach the goal of testing different engine types, intake system design should be modified. The Intake system must be modified and well designed to be capable for a wide range of mass flow to avoid disturbances in intake which causes problems in aerodynamic calibration and uncertainty in thrust measurement. Screens, baffles, vanes, bell-mouths and other treatments must be matched to intake design. In this way, more time and cost should be spent in design and construction steps of multi-engine test cell comparing with the test cell for one family of engines.

4.3 TEST CHAMBER

A perfect test chamber would contain nothing to cause airflow distortion. It would have frictionless surfaces and contain idealized air flows with no viscous effects and

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no vortex formation or exhaust gas re-ingestion. Frictionless flow is not possible, but careful design can prevent ingestion of exhaust gases and minimize vortex formation. Some important parameter that must be considered in design of test chamber are mentioned below:

- Aspect ratio of the test chamber,
- mass flow of air through the engine,
- cell bypass ratio

Aspect ratio of the test cell is the ratio of the cross section area of the chamber to the intake area of the engine. Any thrust correction required will depend strongly on the secondary air speeds through the test chamber. To minimize the correction to be applied to measured values, these secondary air speeds should be kept as low as possible. Generally, this area will be of a sufficient cross section so that the air velocity approaching the engine inlet will not exceed approximately 50 feet per second (fps).

Secondary air velocities cause uneven static pressure distributions along the length of the engine surfaces which give rise to unbalanced forces on the test frame and therefore on the measured thrust. The biggest effect is in the region of the engine nozzle exhaust because here the air velocities are highest as the air is drawn into the augmentor.

Conclusion:

The test chamber must be redesigned to be capable for wide range of mass flow. The TCF increased as the aspect ratio of the test cell decreases. So using bigger engine (cf6) in a constructed test cell causes more TCF and problems in aerodynamic calibration and uncertainty in thrust measurement. In this way, more time and cost should be spent in design and construction steps of multi-engine test cell comparing with the test cell for one family of engines.

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4.4 EXHAUST SYSTEM

The exhaust system of a test cell has great effect on engine performance of a gas turbine because the exhaust system controls back pressure on the engine, and secondary airflow through the test chamber. The exhaust system is also important to reduce the noise emitted into the atmosphere from the test cell.

4.4.1 SECONDARY AIR FLOW

When testing in an enclosed sea-level test bed, the ejector action of the jet stream from the engine exhaust as it enters the augmenter (exhaust collector) induces a flow of secondary air through the cell. The secondary air flow through the intake stack and the test-room and the temperature is very close to the ambient temperature outside the test-bench. This secondary flow rises aerodynamic effects which causes forces to act on the thrust frame which increases errors in the measurement of thrust.

The three main effects are:

- A force on the inlet bell-mouth due to the air entering predominantly from the forward direction;
- A force on the nozzle due to the reduced external static pressure caused by the increased velocity of the secondary air as it approaches the exhaust collector;
- A drag force on the framework supporting the engine due to the secondary air flowing over it.

Increasing of a secondary air flow will generate a more static pressure gradient between engine inlet and nozzle exhaust; this will require a larger thrust correction factor (TCF), which is undesirable.

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The test-bed's exhaust stack must also be equipped with acoustic panels. In order to achieve an acceptable mechanical durability of these panels the exhaust gases' temperature must remain below 400 °C. The secondary air flow plays a very important role in cooling of engine exhaust gas flow.

4.4.2 AERODYNAMIC CONSIDERATION

The velocity and temperature of the gases ejected by the engine depend on the type of engine and the throttle-setting. The outlet section of the jet exhaust duct must be correctly positioned in relation to the entrance to the Augmenter tube. In the convergent intake of the Augmenter tube the secondary air's velocity is augmented and the pressure lowered. If the turbofan outlet section located much closed the convergent intake of an augmentor tube, the thrust will not be measured correctly for the following reasons.

The outer envelope of the exhaust duct and nozzle is in a zone of low pressure and high free stream velocities. Friction and pressure forces exerted in opposite direction of the thrust appear on the outer envelope of the exhaust duct. The pressure on the engine outlet section is lower than that in the test-room. This alters the thrust developed by the engine due to the influence on the equilibrium running point for a non-saturated nozzle.

The outlet section of the engine nozzle must be upstream of the inlet section of the augmentor tube; the recommended distance is one engine nozzle diameter. The volume flow is reduced as much as the distance is lowered so the cell bypass ratio can be changed through the positioning of the engine in relation to the augmentor tube. The jet noise radiates through the augmentor inlet and if the distance between engine nozzle and ejector increases the noise level in the test-room rises. Due to the secondary

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airflow through the test-tunnel there can be unacceptable negative pressure gradient along the engine outer surfaces from in- to outlet.

A larger engine nozzle to Augmenter distance increases noise produced in the test chamber, on the other hand the thrust correction factor (TCF) increases as the distance between the engine nozzle and Augmenter decreases. So it is required to find the optimum condition as a trade-off that is not desirable for target engine (CFM56). Also changes in velocity of the gases in the exhaust, and consequently in the quantity of the secondary air flow, causes change in the size of the augmenter tube. An increase in the exhaust flow temperature has a negative effect in the performance of the ejector pump effect.

Conclusion:

There are some important consideration to design of the exhaust system for a test cell providing test wide range of engine thrust. Some of these issues are mentioned below:

- The engine with higher mass flow needs to more quantity of secondary air flow to keeping the temperature lower than the required level. In this particular case, the mass flow ratio of CF6 engine to CFM 56 engine is more than 1.8, so large change in the quantity of the secondary air flow is needed or we have to change cooling system from air cooling type to the water cooling type or design a test-cell with two air-intakes.
- Having a large engine nozzle to Augmenter distance increases noise produced in the test chamber, on the other hand the thrust correction factor (TCF) increases as the distance between the engine nozzle and Augmenter decreases. So it is required to find the optimum condition as a trade-off that is not desirable for target engine (CFM56).

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- Also changes in velocity of the gases in the exhaust, and consequently in the quantity of the secondary air flow, causes change in the size of the augments tube.
- An increase in the exhaust flow temperature has a negative effect in the performance of the ejector pump effect.

As a result in this way, more time and cost should be spent in design and construction steps of multi-engine test cell comparing with the test cell for one family of engines.

4.5 SUPERSTRUCTURE

The superstructure is designed to withstand the total weight of trust stand, engine and test equipment connected to it with an adequate safety factor. Furthermore in design and manufacturing of the superstructure, its resistance to vibration and resonance should be considered.



Figure 4-5: Superstructure

Conclusion:

For a multi-engine test cell the superstructure should be designed considering enough strength for withstanding the engine and its equipment with maximum weight and

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vibration. In this way, the used material and manufacturing procedure will become more complicated and spend more time and cost in compare with the test cell super-structure for one engine family.

4.6 ENGINE ADAPTER

The adapter is designed to quickly lock the engine into position and allow the engine to receive all necessary services for its testing. The adapter, once mounted in the thrust stand, replicates the way the engine performs on-wing. The adapter comes complete with instrumentation, wiring, and engine service connections.

The adapter is sized by taking the following load scenarios into consideration:

- Fan Blade Out (FBO) test,
- Nominal operating conditions,
- Validation test.

It resists to the incidental strain without any damage.



Figure 4-6: Engine adapter configuration

Conclusion:

The adapter design should be performed based on engine type and thrust level, current adapter (CFM56 family) is allocated to maximum loading due to engine thrust – 150

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kN, So different adapters should be dedicated for different engines. In this way, more time and cost should be spent to design and prepare the appropriated adapters for multi-engine tests.

4.7 THRUST MEASUREMENT SYSTEM

The thrust stand assembly will consist of the following basic components:

- Upper (fixed) assembly that interfaces with the superstructure.
- Lower (floating) assembly that is suspended from the upper assembly by flexure plates. The bottom of the lower assembly provides a neutral docking frame interface to accommodate engine test adapters. The lower assembly incorporates an adapter locking system.
- Docking Frame that provides a neutral interface for bolt-on engine interface adapter that will be provided by others.

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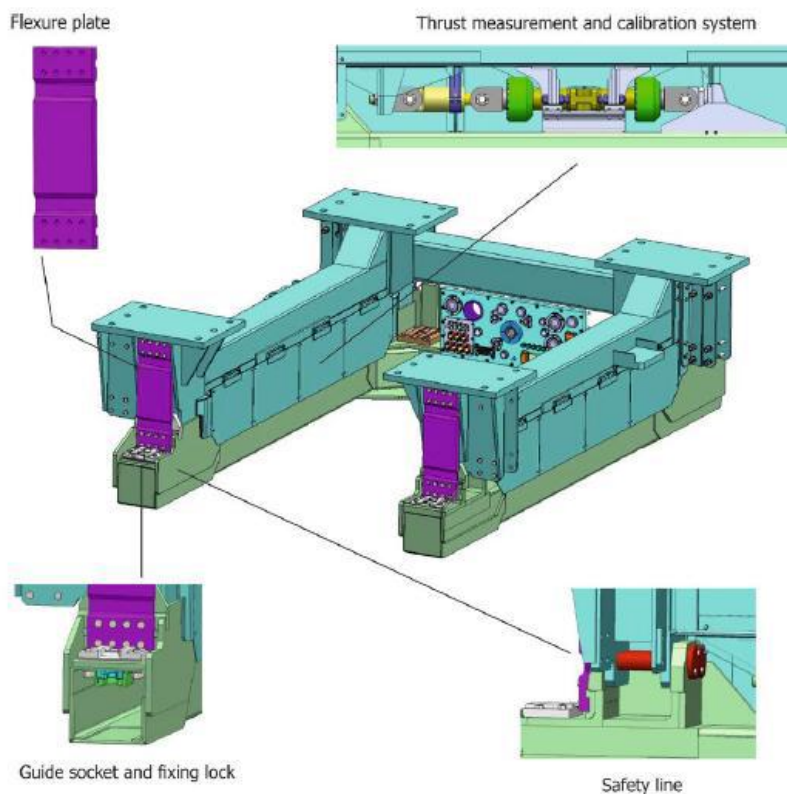


Figure 4-7: Detail of thrust stand

The thrust stand consists of two major assemblies, the fixed frame and the floating (thrust) frame. The thrust stand upper assembly is rigidly bolted to the superstructure. When an engine under test is producing thrust, the floating frame moves longitudinally a few thousandths of an inch and compresses the load cells. The flexure plates which withstand significant vertical loads produced by static engine thrust or engine failure offer negligible resistance to this movement, ensuring that the load cells accurately measure engine thrust. The thrust stand design will consider fatigue stresses imposed on the mechanical elements and structures due to vibration generated as part of normal test activities.

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By changing of engine type some considerations in construction would be met. For a multi-engine test cell the thrust stand and the load cells should be designed considering enough strength for withstanding the engine and its equipment with maximum weight and maximum thrust. The flexure plates are interface between fixed and floating parts of thrust stand. The thrust frame is designed with sufficient stiffness so the weight of the docking frame and engine has minimal influence on the thrust reading from any load cell.

By increasing weigh of engine (and consequently its related components such as a dedicated adapter) the resistance to deflection of service lines that transition from the fixed stand to the floating members of the installation, avoid the force transferring to load cell correctly. As well as capacity of these parts to endure much more weights. Consideration of all potential tare forces (static weight) must be made during design. Tares must be identified, understood, and quantified for data correction.

Conclusion:

For a multi-engine test cell the thrust stand and included load cells should be designed considering enough strength for withstanding the engine and its equipment with maximum weight and maximum thrust, at the same time total weight must have minimum effects on flexure plats functionality and consequently thrust measurement. In this way, the used material and manufacturing procedure of equipment interfacing fixed and floating parts of thrust stand will become more complicated and spend more time and cost in compare with the test cell thrust stand for one engine family.

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4.8 ENGINE HANDLING AND TRANSPORTATION

A wheeled cart will be provided to handle the engine & adapter between the test room and the preparation shop. An engine transport cart is designed for transporting engines, a fan engine, adapter, and test nacelle.

Conclusion:

For a multi-engine test cell with considering the weight differences of various types of engine, the transportation system should be designed with the maximum capability of engine handling with spending more time and cost in compare with the test cell transportation system for one engine family.

4.9 LIFTING PLATFORM

The engine under test is mounted from overhead positioned on cell centerline. The purpose of the work platform is to raise mechanics and their equipment to a convenient working height to perform checks and routine maintenance on the engine.

The elevation platform shall be designed according to the effective labor protection and safety rules.



Figure 4-8: Lifting platforms

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Current lifting platform: 8 ton, 5500 * 9950 mm;

Table 4-1 shows the weight and dimension differences between two different types of engines.

Table 4-1: CFM56 and CF6 weight comparison

	Weight (kg)	Length (mm)	Diameter (mm)
CFM56	1951	2600	1828
CF6	4472	4270	2690

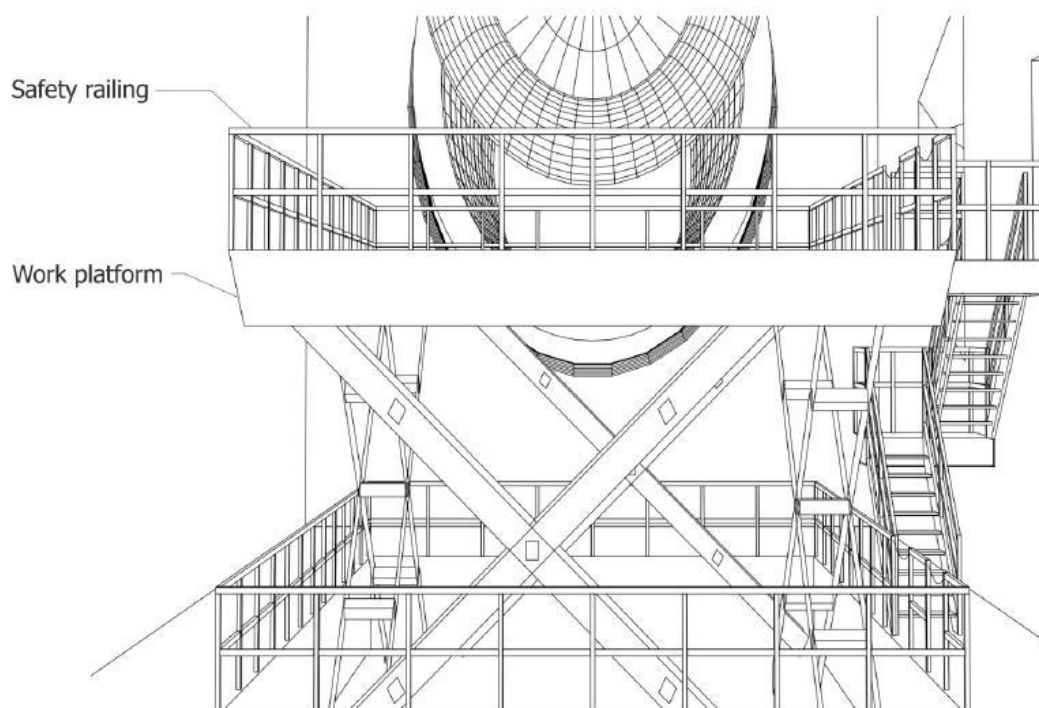


Figure 4-9. Overall view of lifting system

Conclusion:

For a multi-engine test cell with considering the weight differences of various types of engine and its adapter, the lifting system should be designed with the maximum capability of engine handling with spending more time and cost in compare with the test cell lifting system for one engine family, as well as having appropriate size.

4.10 VIBRATION DAMPING SYSTEM

Some of the main sources of vibration in test cells is:

- The amount of volumetric flow and the possible vorticities generated in some areas.
- The speed of engine rotors.
- The amount of engine thrust.

The test cell components such as superstructure and thrust stand should be designed considering the maximum level of mass flow which produce high vibrations and fatigue stresses.

Conclusion:

The natural frequency of the thrust stand with attached docking frame must be outside the engine's excitation range which may cause designing a vibration damping system which is adapted with changing the engine type and to isolate the maximum level of vibration with sufficient effectiveness.

In this way, more time and cost should be spent in design and construction of multi-engine test cell vibration damping system comparing with the test cell for one family of engines.

4.11 AUGMENTER AND BLAST BASKET

The flow in the exhaust stack and between the acoustic panels must be as uniform as possible. With a simple cylindrical augmenter tube, without any turning vanes and blast baskets, engine exhaust gases turn to the cell exhaust stack sharply which causes massive separation on stack wall and consequently cross area of the exhaust would be reduced at this section. Decreasing in cross section makes flow accelerated, so the

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high velocities in this non uniform flow reduce the acoustic treatment's life and generate noise at the outlet.

To achieve a fully mixed flow in a simple cylindrical detuner (augmenter tube) would require an excessively long detuner, and thus a 'blast basket' is used to enhance mixing and reduce length.

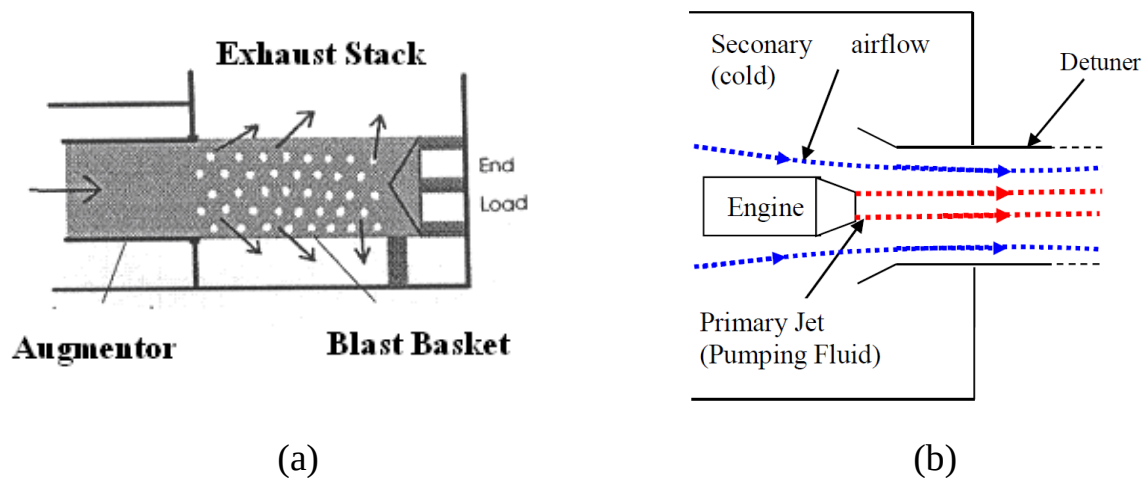


Figure 4-10: Test cell exhaust system: a) Blast Basket. b) Detuner and mixed flow

4.11.1 BLAST BASKET

A blast basket is simply a cylinder with thousands of holes drilled into it. The holes should not be too small in order to avoid clogging by carbon deposits. Blast baskets also redirect the flow from horizontal to vertical ejection through a stack and improve flow uniformity in the exhaust stack. The final exhausted mass flow is fully mixed, low velocity, low temperature flow. Blast basket are used for exhaust's low frequency noise, when it changes into a high frequency noise which can be attenuated more easily by the acoustic material.

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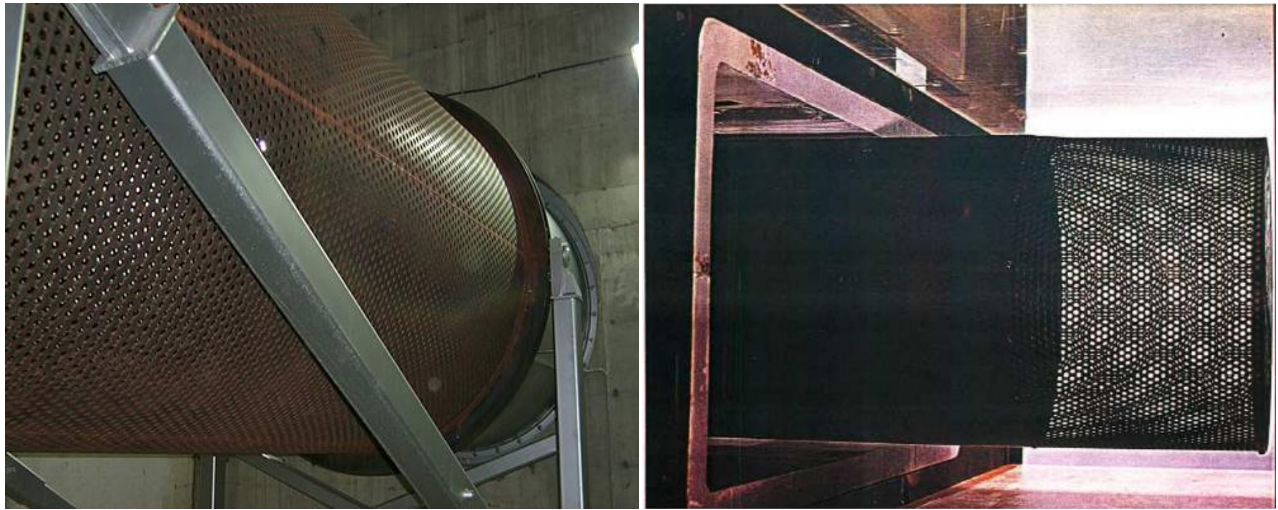


Figure 4-11: Blast basket

4.11.2 AUGMENTER TUBE (DETUNER)

The exhaust system of the test cell includes a detuner which collects engine exhaust gases and secondary air and ejects them to atmosphere.



Figure 4-12: Augmenter tube

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High energy exhaust gases flow at high velocity while the secondary airflow has low energy and velocity. The mixing and interaction between these two airflows at the entrance to the detuner produce an ejector pump effect which is both important and quite complex. However, the ejector effect will have an effect on the pressure at the engine inlet possibly altering its performance.

The pumping efficiency depends on the augments geometry:

- L/D : length/diameter ratio
- A_n/A_{aug} : Ratio of the nozzle exhaust cross-section to augments tube cross section.
- X/D_n : Ratio of the distance between engine nozzle exhaust and augments inlet to nozzle diameter.

The augments tube's diameter is usually 3 times that of the engine nozzle exit diameter and A_n/A_{aug} is about 1/9. To obtain an acceptable airflow pumping efficiency the length/diameter ratio should be greater than 8. The outlet section of the engine nozzle must be upstream of the inlet section of the augments tube; the recommended distance is one engine nozzle diameter. It should be noted that the cell bypass ratio is influenced by the distance between engine nozzle exhaust and augments inlet, ratio of the diameters of the nozzle and augments tube and primary flow pressure and temperature.

The mixing of the two streams reduces the temperature and velocity of the primary flow which are critical factors in the reduction of both cell cost and noise pollution. The noise of a jet of air is directly related to the velocity of the jet hence reducing the velocity of the jet will reduce noise pollution generated by the test cell. The reduction of exhaust jet temperature reduces the need for complex designs and use of heat resistant materials in the exhaust system, thus reducing cost (and size) of the test cell.

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Conclusion:

Augmenter tube length, diameter and blast basket diameter should be designed in order to reach the goal of testing different engine types. In this way, more time and cost should be spent in design and construction steps of multi-engine test cell comparing with the test cell for one family of engines.

4.12 BELLMOUTH

The bell-mouth is a smooth convergent duct allows the maximum amount of air to be drawn into the engine with the minimum pressure loss. In the design of this facility special attention is dedicated to avoid the influence of ambient disturbances as the wind which could seriously affect the test results. For this reason the bell-mouth can also be supported by a large mesh screen fitted around the engine.

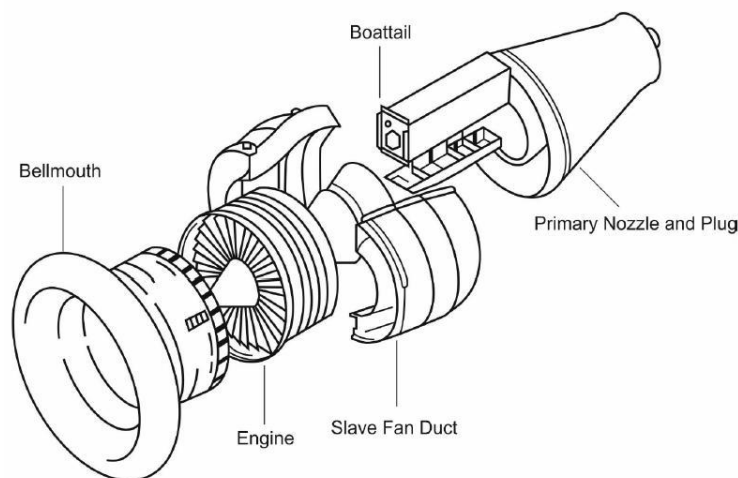


Figure 4-13: Test Nacelle equipment

Conclusion:

Specific bell-mouth must be provided for each engine with different dimensions. In this way, more time and cost should be spent in different bell-mouth design and construction of multi-engine test cell comparing with the test cell for one family of engines.

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5. AUXILIARY SYSTEMS

In this section, the main components of test cell auxiliary systems and their design considerations are described. Also, the feasibility study of auxiliary systems in multi-engine test cell has been performed and compared with the test cell for one family engine testing in terms of design complexity, time and cost.

5.1 FUEL SYSTEM

The purpose of the fuel system is to safely supply fuel to the engine under test at pressures' and flows' that fulfill all test requirements. There is proposing to supply fuel metering and conditioning packages that filters, controls the pressure, monitors the flow and provides fuel for safe engine shut down in case of failure of the fuel supply system.

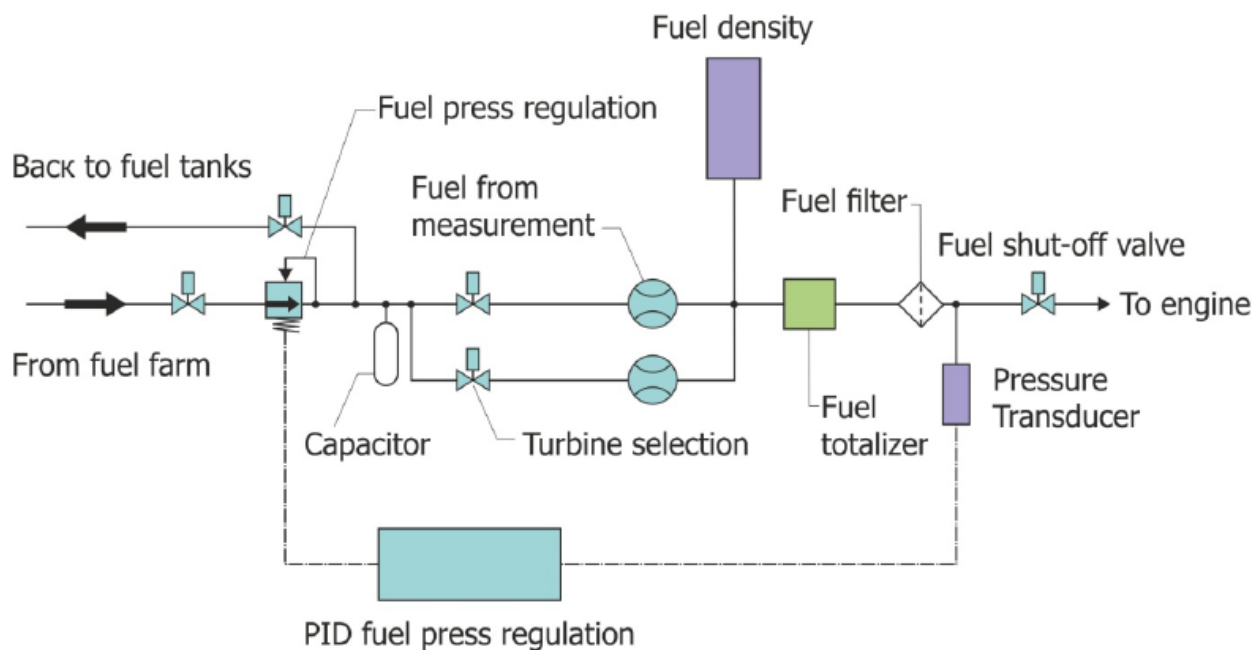


Figure 5-1. Example of fuel metering and conditioning

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Conclusion:

Specific system must be provided for each engine with different characteristics such as type of fuel, required fuel capacity (pumps, heat exchangers, filters, piping, and valves) and control systems. In other case, one system could be design for a multi-engine test cell considering the maximum capacity but it should be modified for each engine because of little differences in access points of components. In both cases, more time and cost should be spent in fuel system design and construction of multi-engine test cell comparing with the test cell for one family of engines.

5.2 OIL FILLING SYSTEM

The lube oil filling system will make it possible to fill the engine oil tank remotely with the engine installed in the thrust frame.

Specific system must be provided for each engine with different characteristics such as type of oil (to resistance to high temperature, prevent corrosion and changes in solution chemistry), required oil capacity (pumps, heat exchangers, filters, piping, valves) and control systems.

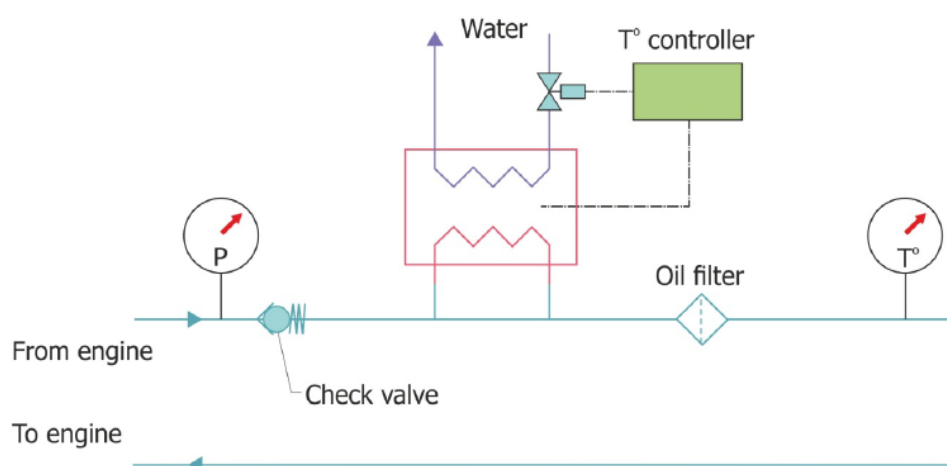


Figure 5-2. Example of oil filling system

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Conclusion:

Specific system must be provided for each engine with different characteristics such as type of oil, required oil capacity and control systems. In other case, one system could be design for a multi-engine test cell considering the maximum capacity but it should be modified for each engine because of little differences in access points of components. In both cases, more time and cost should be spent in design and construction steps of multi-engine test cell comparing with the test cell for one family of engines.

5.3 SUPPLY SYSTEM CAPABILITIES

This item included power supply, water supply, air supply etc. will be increased with increasing in thrust value and construction a test cell with the capability of testing the different types of engine. In this way, more time and cost should be spent in design and construction of supply systems in multi-engine test cell comparing with the test cell for one family of engines.

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6. DAQ & CONTROL SYSTEM

In this section, the main components of DAQ & control system in multi-engine test cell and in the test cell for one family engine are compared with each other in terms of design complexity, time and cost.

6.1 CONTROL SYSTEM

Some components of current control system are suggested based on only CFM56 family and must be changed if another types of engine will be intended. These components include both hardware and software as following:

- An ARINC 429 module is required to communicate with engines that have an Electronic Engine Control (EEC) unit. This includes the CFM56-5 and CFM56-7 engine types, so an extra module or modified one should be re-placed for CF6 family engines.
- Current throttle system is specified for the CFM56-5 and -7 consists of dual resolver signal inputs to the engine EEC. So an extra throttle system should be provided for CF6 family engines.
- The system will be delivered with initial engine specific software for test bench commissioning with a CFM56 engine. The software contains the full settings to perform the full test of the CFM56 engines, in accordance with the official documents from the OEM.

Conclusion:

Some hardware and software components must be changed or redesigned to follow the target of testing both CFM56 and CF6 family. As software will be programmed based on engine documents, two different softwares will be required for two types of engines (CFM56 and CF6 families. In this way, more time and cost should be spent in

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control system design and construction of multi-engine test cell comparing with the test cell for one family of engines.

6.2 ACCURACY OF MEASUREMENT

Type of the engine and its thrust class have huge effects on thrust stand components' characteristics, capabilities and setups, and also uncertainty of measured values. Some the most important consideration listed below.

- The design and construction of turbofan/jet engine test facilities will influence engine operation and performance. Hardware installed on engines, such as bell-mouths, thrust reversers, exhaust nozzles, noise suppressers, cowlings, and quick engine change (QEC) components, can alter engine performance and influence test results. Therefore, accurate turbofan/jet engine performance evaluation requires that the influences of both the test facility and the engine, including its installed equipment, be known. Test cell correlation testing is performed to determine the effect of these items on the performance of an engine relative to the baseline performance of that engine. When a test cell is correlated for a specific engine model, the correlation may also be valid for derivative models of engines having lower thrust but using the same config and same hardware design. So obviously correlations must be investigated for each types of the engines separately which leads to extra correlation testing, as well as recorrelation for new test cell design for a specific type of engine(new design for CFM56 family). Another item that makes recorrelation necessary is modifications have been made to existing data acquisition or data reduction systems when these modifications are introduced in the test cell in a manner that could affect engine performance evaluation;

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- In addition as mentioned before, the systems such as inlet and exhaust should be modified and required sensors should be added to the system in order to reach the way to recalculate the thrust correction factors as precise as possible. In this way, more time and cost should be spent in multi-engine test cell comparing with the test cell for one family of engines.
- As Maximum thrust will increase from 45000 lbf to 90000 lbf, higher range load cells and consequently calibration system (master calibration load cell and hydraulic force generator) must be prepared. It should be mentioned that using high range load cell for low range force measurement increases total errors.
- Errors accounted for by the system calibration are primarily those resulting from system static weight loads. So **by changing weight of the engine and its attached components, the calibration of the thrust stand should be repeated.**
- Generally, the aerodynamic calibration should be modified for each engine in order to reduce the measurement errors. In this way, more time and cost should be spent in design of multi-engine test cell comparing with the test cell for one family of engines.

Conclusion

Since the engine and its associated components become changed, total weight changes and must be considered in new calibration. All thrust correlations are specified for one type of engine, so separated correlation factors and consequently extra set up and measurements will be needed for each types of engine.

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7. MISCELLANY

In this section, the main components of test cell miscellany and their design considerations are described. Also, the feasibility study of miscellany in multi-engine test cell has been performed and compared with the test cell for one family engine testing in terms of design complexity, time and cost.

7.1 METROLOGICAL STATION

The meteorological station will measure parameters of the ambient environment during engine tests required for calculations of corrected engine test results. This system for a multi-engine test cell is the same as the system in a test cell for one engine family testing.

7.2 FIRE EXTINGUISHING SYSTEM

7.2.1 FIRE PROTECTION SYSTEMS FOR THE TEST STAND AND EXPOSED AREAS

The required water for fire protection will be made available for connection to the proposed test stand deluge system, with a minimum flow and pressure capability of 230 m³/h at a pressure of 10 bar. As these calculations were made of based on current dimension and plan of construction, it's obvious that capacity of fire distinguish system should be recalculate.

7.2.2 UNDER-COWL FIRE EXTINGUISHING SYSTEM

The test adapter is equipped with gaseous discharge fire protection, multiple zone core temperature measurement, fire shielding, fuel isolation, and pressure relief of fire protection. As all these components would be accommodated in a dedicated adapter, they must be considered separately for each adapters (engine types).

7.3 CCTV & INTERCOM

Video surveillance with digital recording shall provide visual check inside:

- Test cell (engine, doors, test equipment, etc.);
- Control room;
- Auxiliary rooms (test equipment and technological systems).

In order to allow that, video color cameras will be installed. These cameras will be also equipped with a motorized zoom allowing looking onto details of engine. All the controls will be carried out from a remote control unite installed in the control room. Views will be displayed on a color screen integrated into the control console.

This item will be unchanged with increasing in thrust value and construction a test cell with the capability of testing the different types of engine.

An intercom circuit will be installed to communicate easily between the test cell and the control room. Communications will be hand free type. It will not be necessary to maintain a switch “on” during communication. For a better listening, loudspeakers will be installed in the test area.

This system for a multi-engine test cell is the same as the system in a test cell for one engine family testing.

8. ASSESSMENT OF ENGINE TYPES FOR IRAN AND INTERNATIONAL FLEET

Engine test cell development and construction is a critical step in modernisation of infrastructures of aviation industry. It adds a powerful capability to the spectrum of national assets to guarantee proper and continuous delivery of commercial air transportation services and also keep the fleet of military aircrafts in full operational status in a timely manner.

As a complex project with high-technology aspects, development and construction of an engine test cell is a challenging job to do. One of the main steps in the conceptual definition of the project is the process of freezing a value for the thrust level to be measured in this cell. There is no well-defined method or a dedicated numerical formula for calculating this amount. This parameter must be defined as a convergence of design philosophy, environmental limitations, operational requirement and executive priorities. Taking all these items under the same umbrella is a hard task to conduct.

In this document a thorough analysis of the different aspects dealing with engine test cell development and construction project has been done covering current and future fleet size of Iran, military aircrafts and their engines serving in the armed forces of IRI, regional as well as international trends and opportunities and the trends governing future of air engine industry and market. Finally a method is presented to calculate the priority for the thrust level to be defined as the operational margin of the test cell.

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8.1 CURRENT CIVIL JET AIRLINERS FLEET OF IRI

Iran has over 330 airports, and more than 120 of which had paved runways. Of those, 41 (40 paved and one unpaved) had runways 3,000 meters or longer. International airports of Iran are actively operating jet airliners from both domestic and international airlines. Iran's commercial airline companies suffered from years of sanctions which prevented them from buying new aircraft, or even making repairs to existing aircraft with manufacturer-produced parts. But they keep delivering services to the growing population among these challenging environment.

Reviewing the history of commercial aviation growth in Iran, one can clearly understand the strong role of the government which was getting some supports by introducing the new private airlines in the country since recent 2 decades. In the case of operation the main problem of the fleet is the age of the aircrafts. The sanctions made the job harder for the airlines. Engines as the heart of propulsion system for the aircrafts need careful planning in the case of maintenance and overhaul. Figure 8-1 depicts the total jet airliners available in Iran both operational and grounded as of August 2017 and the numbers of corresponding engines.

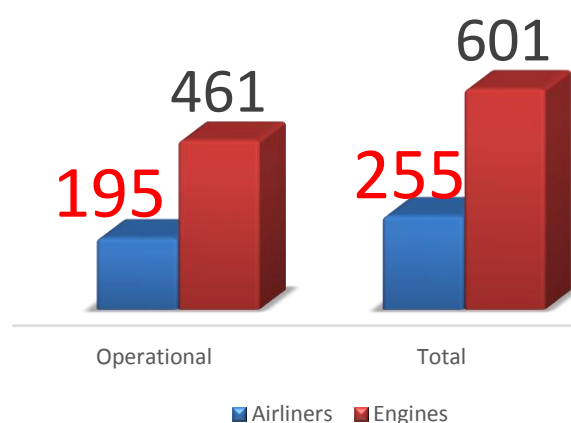


Figure 8-1: Number of Operational and Grounded Jet Airliners and Engines

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Take a glance at the chart above may be disappointing, but reminding the circumstance in which Iran aviation industry has been working for about 4 decades will lead us to the brighter side of the story. Turning it into a business frame of work and focusing on it as a market, it will be attractive in case of statistics. More than the total figure, the distribution of these aircrafts and engines through the network of the operators is an important factor to make the potential market more robust. Figure 8-2 listed the statistics for the total and operational jet airliners and their engines for all the active airlines in the country.

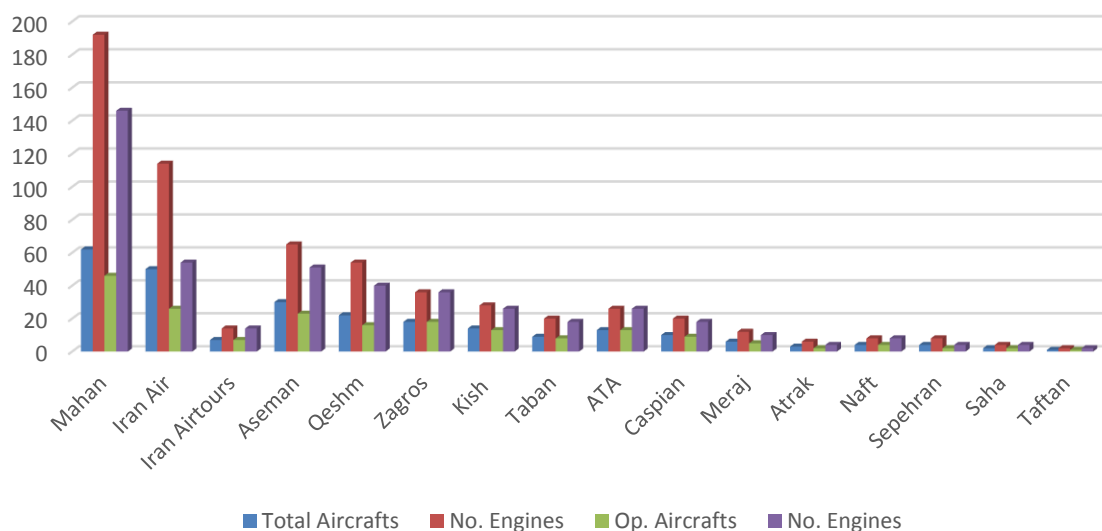


Figure 8-2: Number of Total/Operational Jet Airliners and Engines for the Airlines

The names of 16 different airlines with different fleet sizes in the chart can be seen. It is important to notice that most of these airlines are private and they decide on their strategies based on their own objectives and priorities. Near one thousands engines in the country and no test facility, shed the light for who is going to establish an engine test cell here. Taking this step with confidence for the size of market, the main consideration will be the sizing of the test cell. Talking operationally now it's the time

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to freeze the thrust range. Figure 8-3 presents more details on the thrust range of the engines mounted on both operational and grounded aircrafts.

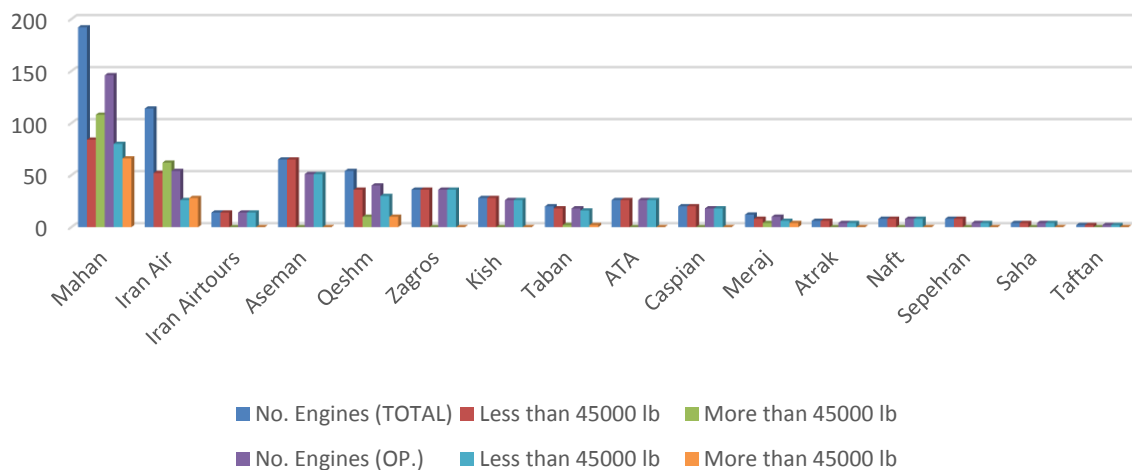
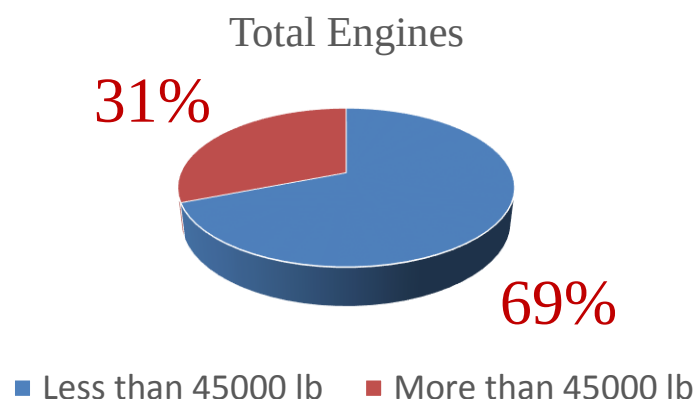


Figure 8-3 : Number of Total/Operational Jet Airlines and Engines for the Airlines

An overall look at the chart presented in Figure 8-3 helps us find out the majority of the engines have the thrust less than 45000 lbf. To make a more effective feeling on this Figure 8-4 summarises these data in a more compact appearance.



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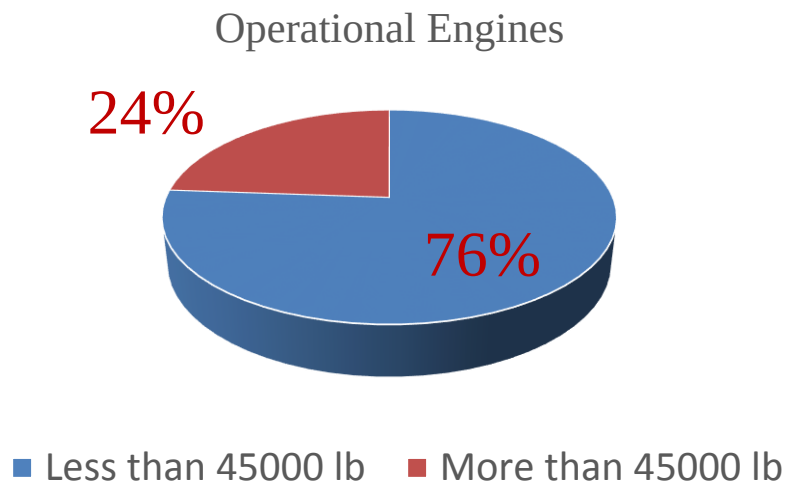


Figure 8-4: Comparison of Total and Operational Engines with Different Thrusts

The collective data analysis presented in the figure above confirms that the majority of the engines provide power for Iranian jet airliners produce thrusts less than 45000 lbf. It is more interesting when taking into account that the same results has been achieved comparing the engines for the operational aircrafts of each individual airline. For 11 airlines the full fleet of the airliners are powered by the engines with the thrust below 45000 lbf. For the remaining 5 airlines the percentage of the operational engines with the thrust less than 45000 lbf is considerable and above 70 percent.

8.2 ORDERS AND FUTURE DELIVERIES

Global aircraft makers are keen to turn towards Iran ever since western sanctions on the country were lifted recently, as the country looks to replace its ageing fleet of about 250 commercial aircraft of which almost half are grounded for want of spares. Iran's 16 airlines, including state-owned IranAir, have a combined capacity to fly 22 million passengers per annum, according to aviation research body Centre for Aviation (CAPA). Up to now four of Iranian airlines completed their orders of signed MoU with Airbus and Boeing to buy new jet airliners. Figure 8-5 not only shows the number of aircrafts to be delivered to Iran for the coming years but also depicts the engines according to their thrust.

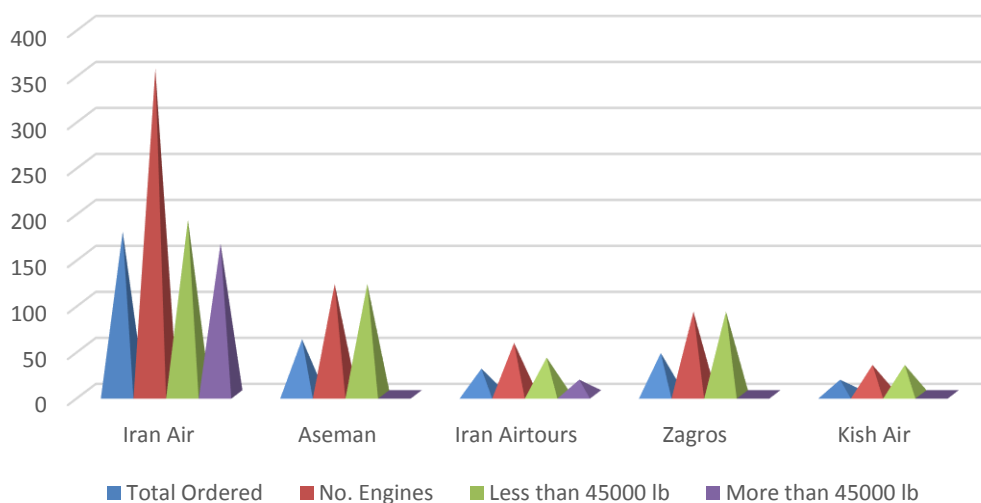


Figure 8-5: Total Jet Airliners Ordered by Iranian Airlines

The same conclusion raises by comparing the thrust level of the engines to be mount on future Iranian jet airliners. More than 71% of the engines produce thrust less than 45000 pounds. Amazingly both Aseman and Zagors are acquiring new airliners all with thrust levels less than 45000 lbf.

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8.3 MILITARY AIRCRAFT FLEET OF IRI

Engine test cell is capable of evaluating both civil and military propulsion systems. In previous chapters the current and future fleet of the commercial aviation industry of Iran has been analysed. Doing the same for the military sector is challenging because no official information is available. By the way using credible internet sources the following table summarizes the current fleet of Iran military aircrafts.

Table 8-1: Total IRI Military Aircraft Fleet

F-14 Tomcat	Lockheed Jetstar
Mig-29 Fulcrum	B707
Mirage F-1	B747
Chengdu F-7	Falcon 20
Saeqeh	Il-76
F-4 Phantom	An-74
F-5 Tiger	Chengdu FT-7
Azarazkhsh	Simorgh
Su-24	Su-25

Table 8-1 covers the complete list of IRI military aircrafts including fighters, bombers, transport and trainers all equipped with turbojet/turbofan engines. In the case of power-plants the thrust range for these birds are categorized in Figure 8-6. Again the

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margin of 45000 lbf has been set. Considering the list in Table 8-1 the total number of engines installed on IRI military aircraft fleet is 693.

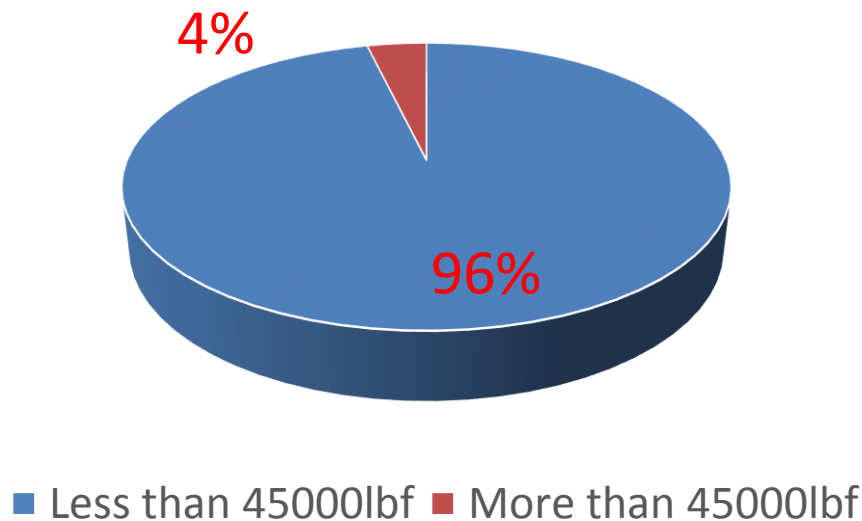


Figure 8-6: Thrust range of engines mounted on IRI military aircraft fleet

The result is obvious and the majority of the engines installed on aircraft serving in the armed forces are producing thrust less than 45000 lbf amount. Considering the plans for modernisation and also acquiring new warplanes in the coming years especially the project of buying Russian Su-30 fighters endorses the evidence as future fleet of military air engines for IRI are categorized as engines with thrust level of less than 45000 lbf.

8.4 INTERNATIONAL TRENDS AND REGIONAL PERSPECTIVE

China and Russia are making their presence felt in the multibillion-dollar aviation market. With demand for single-aisle aircraft expected to grow for years to come, particularly from Asia, some experts think the new firms have a real chance at cracking the market, especially if they have the backing of their national governments. But it won't be easy loyalty to the likes of Airbus and Boeing for one will remain a factor, for sure.

MC-21 aircraft in a few years is intended to compete directly with Boeing's 737 Max 8 and Airbus' A320neo. Irkut was joined by Comac, which is also launching a single aisle plane, the C919. The plane, which is set to carry around 175 passengers, is scheduled to hit the market in 2019.

Airbus predicts that global airlines will purchase 33,070 planes over the next 20 years, worth \$5.2 trillion, while Boeing forecasts the need for 39,620 aircraft worth more than \$5.9 trillion. According to Airbus, 71 percent of the total needed will be single aisle airplanes, the types the Russian and Chinese plane-makers are launching. China will undoubtedly be a big player. The ballooning middle class in the country, now the world's second-largest economy, mean passenger journeys are expected to double over the next 20 years.

To make it easy for Russia and China to find international market for their new jet airliners they both offer one domestic engine and one western engine to be installed on the aircrafts purchased by the customers. Comac selects CFM LEAP for C919 and Irkut offers PW 1431 for MC21. The point is the thrust level of both engines fall below the amount we are emphasizing in this report as 45000 lbf.

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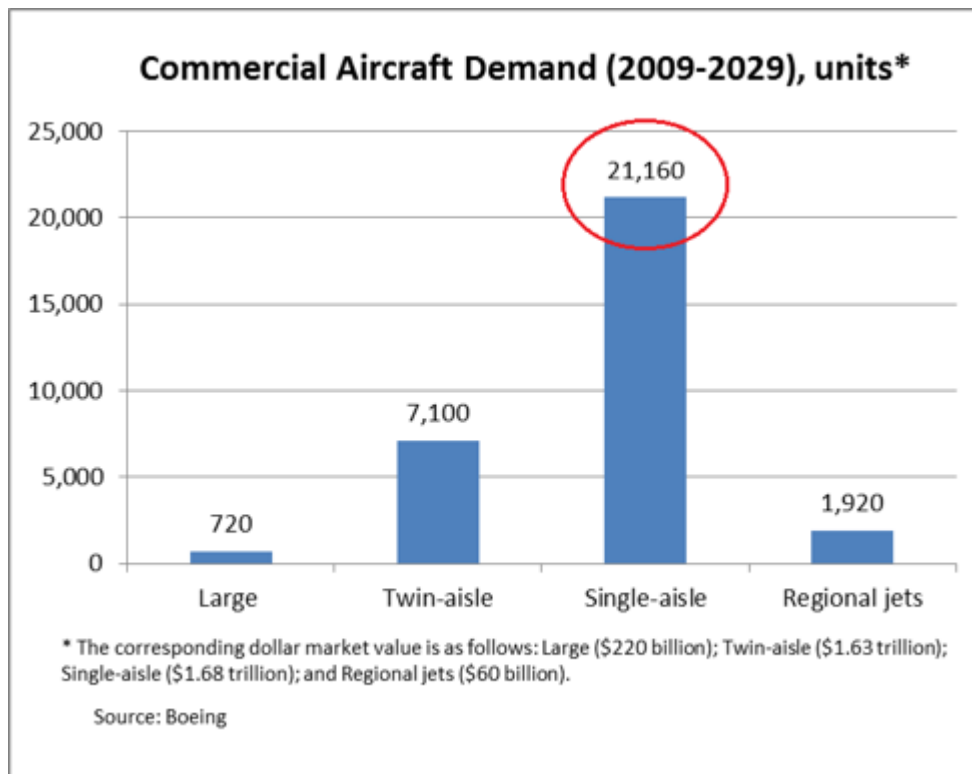


Figure 8-7: Boeing forecast of commercial airliners market

We've just started reviewing international trends by introducing newcomers. Now it's the time to take a look at the main players of the market Boeing and Airbus. The best-seller airliners for both giants of the industry include A320 and B737 families each with a figure of more than 7000 aircrafts sold and a great numbers of backlogs. Forecast for the future is positive. Figure 8-6 depict how Airbus predicts the market for next 20 years.

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Table 8-2: Civil jet aircraft market shares by different regions

	2017 2026	2027 2036	2017 2036	SHARE OF 2017-2036 NEW DELEVERIES
AFRICA	350	700	1050	3%
ASIA-PACIFIC	6140	8140	14280	41%
CIS	340	860	1200	3%
EUROPE	2640	4180	6820	20%
LATIN AMERICA	940	1730	2670	8%
MIDDLE EAST	1260	1270	2530	7%
NORTH AMERICA	2360	3260	5620	16%
FREIGHTERS	410	320	730	2%
WORLD TOTAL	14440	20460	34900	100%

Planning an engine test cell for Iran, the potential market will not be limited only to the country. In the Table 8-2 the rise in CIS and Middle East region is obvious alongside with the overall increase in global scale.

Although UAE has invested in air engine MRO sector for many years, but according to the capacity of the market of the region and the political issues Iran can keep its eyes wide to a considerable share of the regional market in this area.

The commercial aviation market is not only operated by big governmental companies anymore but also the small private players mostly the LCCs are actively acting their roles.

Providing engine test services for airlines, different approaches must be followed facing with the governmental or private companies in the region. The political relations with the neighbours and the overall regional balance will open the windows for high-level negotiations and attract the flag carriers to use such services in Iran. But for the smaller private companies, economy and getting proper services with lower

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costs will be a great motivation. The combination of these strategies will pave the way for a fruitful market ahead.

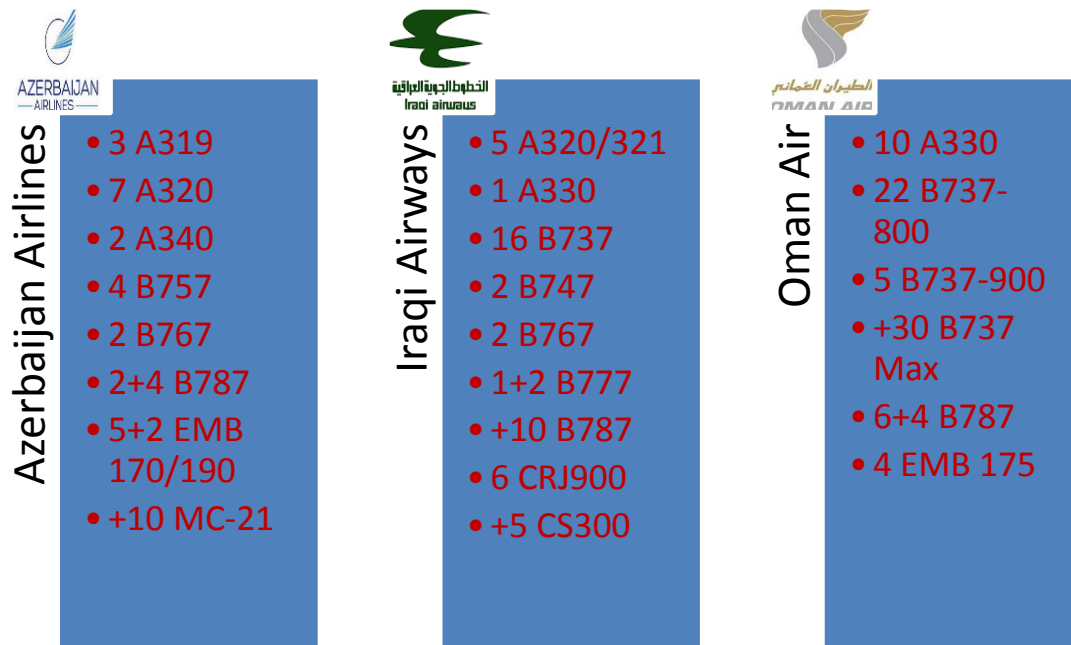


Figure 8-8: Operational and future fleet of selected regional airlines

Above three examples of active regional airlines are addressed with the current fleet and the ordered ones with a plus sign added. An overall view to their fleet shows us the majority of their jet airliners are equipped with engines with thrust lower than 45000 lbf.

The global air transport jet engine fleet will grow by more than 20,000 net new engines by 2025.

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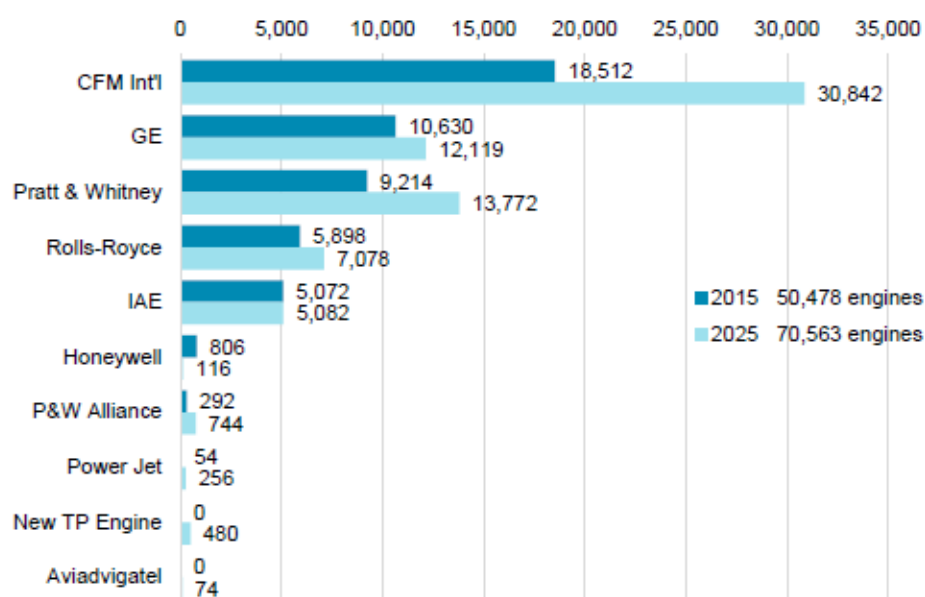


Figure 8-9: Forecast of air engine production and delivery

The highlights:

- Global engine fleet will grow on average 3.4% over the next 20 years
- CFM International fleet has the greatest market share
- CFM56 and LEAP fleet will be a major player and the best seller

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8.5 CONCLUDING REMARKS FOR SIZING AN ENGINE TEST CELL

This document has been prepared in order to address the main factors shape the philosophy of overall sizing process for an engine test cell mostly selection of its range in the framework of thrust. The test cell for aircraft jet engine is the equipment in which performance checks are conducted after maintenance of an engine, and checks the operating condition of each section, adjustment of each part before loading to the aircraft etc., by a trial run of the engine. As this equipment is developing to test the air engines it would be useful to take a look at the portfolio of commercial turbofan engines for wide and narrow body civil jet aircrafts in the picture below:

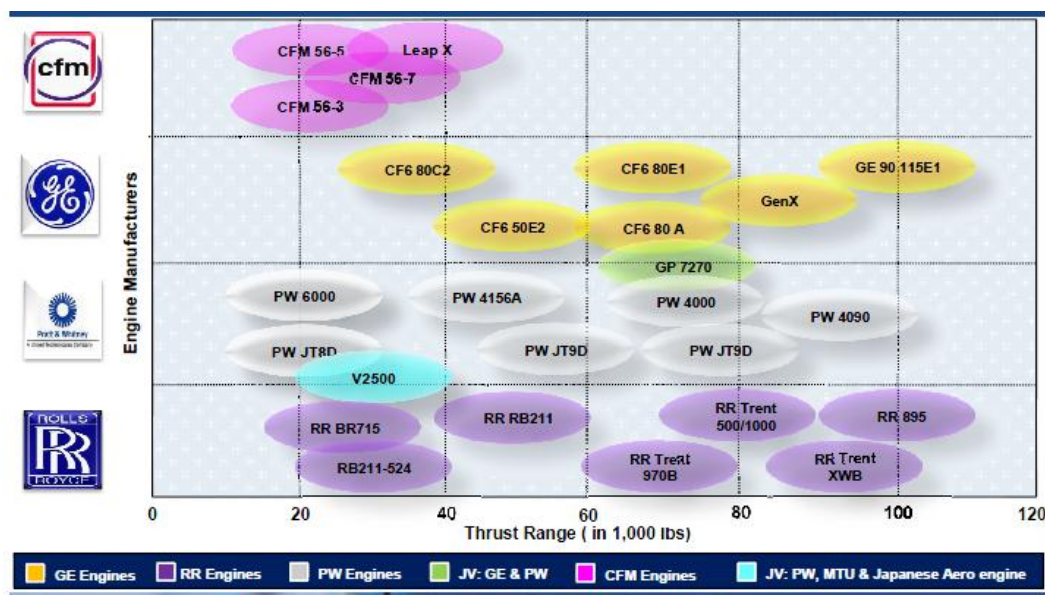


Figure 8-10: Portfolio of commercial turbofan engines for civil jet airliners

The carpet diagram above gives a big picture of what is happening in air engine market. Roughly estimation draw two boundaries for thrust value produced by the

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engines as 20000 lbf and 10000 lbf. As discussed in the previous chapters CFM international is leading this market and according to the forecasts it will be the main supplier of air engine for the next 2 decades in global aviation market. Considering the high production rate for A320 and B737 families and also the newcomers from China and Russia to the market of civil jet airliners, CFM56 air engine family will be the focal point of the market. Accordingly the thrust range of this family which will be followed with the more advanced models of LEAP engines fall below the 45000 lbf. This is the main reason this value has been repeated throughout this report. For sure selecting a thrust range as the working condition for the engine test cell is influenced by many factors. In this document the reality of Iranian commercial aviation industry alongside with the future deliveries and the regional opportunities and the international trends were addressed as the main contributors.



Figure 8-11: Cycle of contributing parameters

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The figure above depicts the main contributors considered for freezing a limit for the thrust range of the engine test cell. These parameters are external factors imposing by the operational environment. To investigate the other thrust ranges, the current and future fleet of air engines for Iran commercial aviation industry has been checked in the figure below.

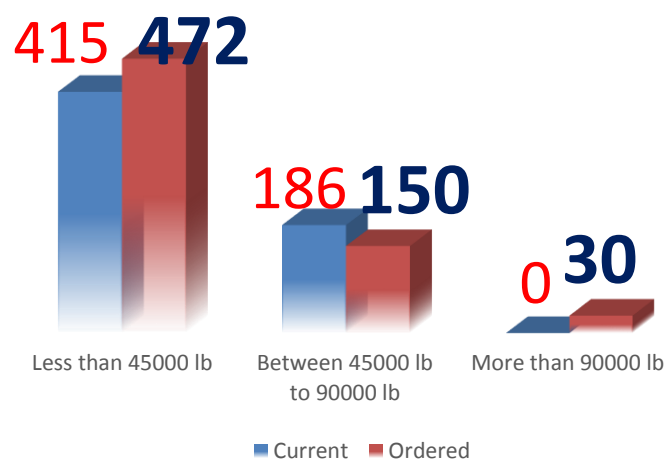


Figure 8-12: Commercial aviation engines categorized in 3 thrust ranges

Analysing the above figure result needs let us widen our options of the thrust range theoretically. But practically it's not rational to choose the ranges more than 45000 lbf because the majority of current and future air engines of the fleet have the thrust less than 45000 lbf.

It's the proper time to change the direction of analysis from a top-to-bottom view into a bottom-to-top one. Up to now different charts and statistics have been presented alongside the expressive analysis to describe the situation of air engine test cell which is going to be developed. In the other world the overall sizing parameter of thrust range were dictated by a market-driven analysis.

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The other side of the coin reflects some more considerations. An engine test cell is a complex high-technology facility which needs special and very careful sizing and designing process to be followed. The cost and the time of conducting such project is very crucial. The smart selection of the main operating parameters like the thrust range is very critical in managing the cost and the time. So it must be selected through a rational procedure considering all the prospects.

Technically setting a value as the limit of thrust level for the test cell must be done in accordance with the availability of the engines to be tested. The level of 45000 lbf has been selected strongly endorsed by market survey. As mentioned both locally and regionally the current and future commercial aviation market is driven by CFM-56 engines. The thrust of this engines fall below the margin of 45000 lbf. Take a look to the list of air engines the other competitor is CF-6. Setting this engine as a reference for sizing the test cell will lead us to the narrowest part of the availability criteria. It means that the cost, time and complexity of the engine test cell development and construction will be increased in favour of just the smallest slice of the available market.

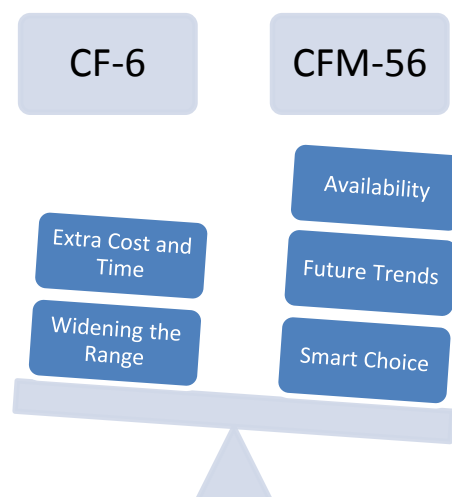


Figure 8-13: Comparison criteria for two main candidates

8.6 DECISION MAKING PROCEDURE BASED ON CONTRIBUTING FACTORS

Decision making is the process of making choices by identifying a decision, gathering information, and assessing alternative resolutions. Using a step-by-step decision-making process can help making more deliberate, thoughtful decisions by organizing relevant information and defining alternatives. This approach increases the chances to choose the most satisfying alternative possible.

The objective of this chapter is to define a process through which the main decision making will be done for sizing the engine test cell by selecting the best thrust range. As the first step the main contributing factors are listed in Figure 8-14.

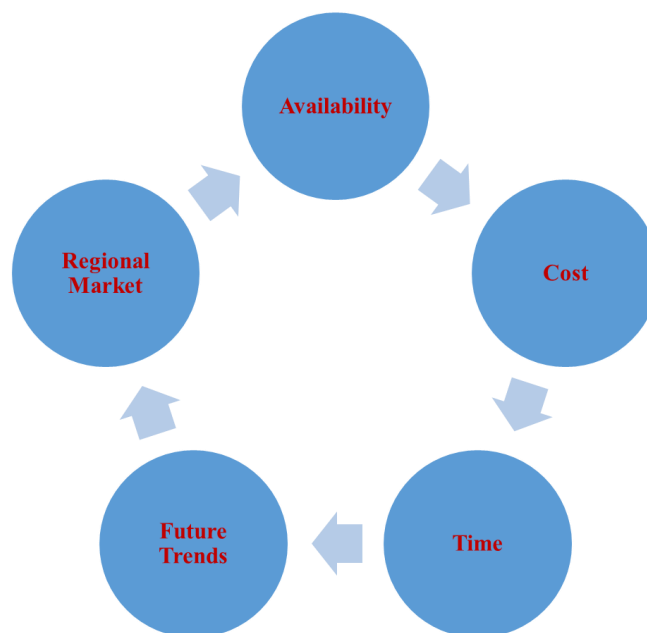


Figure 8-14: The main contributing factors

To define a quantitative criteria for this decision making process the next step will be scoring each factor. The total score is set 100 and according to the expertise and experiences of the team the scores for each factor is listed in Table 8-3. It's necessary to remind that among the 5 main factors listed in Figure 8-14 two of them including

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time and cost are solid factors but the other 3 are combined factors influenced by a set of other sub-factors. It will be addressed in Table 8-3.

Table 8-3: Contributing Factors and their Scores

Factor	Score	Comment
Availability	40	Function of current and future civil fleet, military fleet, number of airlines with the most availability, number of aircrafts with maximum availability
Regional Market	10	Operational fleet of the regional airlines
Future Trends	15	Global trends of production and delivery
Cost	25	Cost of the development and construction project
Time	10	Time of the development and construction project

Among the factors availability is more complicated and itself is a function of other parameters. This factor has been defined in order to make sure the complete coverage of current and future status of operational fleet mostly in the civil aviation sector and also the military aviation. Other aspects of availability in regional and international perspective is addressed via other two factors. The cost and the time is important because they affect the overall decision making process of the whole project, so the interaction between the thrust level margin selection and the scope of cost and time must be investigated.

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Three scenarios are selected for this quantitative analysis including 45000 lbf, 65000 lbf and 90000 lbf. The calculation procedure for scoring starts with comparing each scenario with others through the main factors. Finally the score will be calculated in accordance with a technique known as max-min method. In this method the top list candidate score the highest and the bottom list score zero. In this study because of the limitation in the selection domain including just three cases, the scoring method has been modified and the complete score has been allocated to the top list but other candidates get their score in a linear correlation according to the top list. The result has been presented in Figure 8-15.

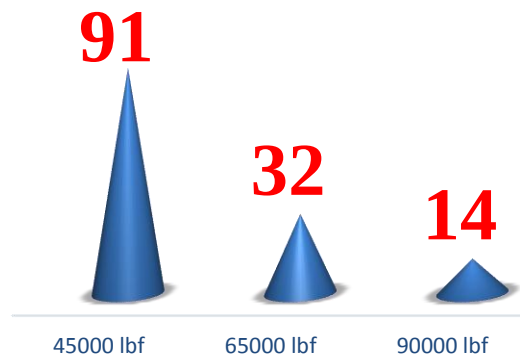


Figure 8-15: Scoring based on the contributing factors out of 100

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9. CONCLUSION

In this section from the point of view of facilities and equipment, the feasibility study and comparison of CFM56 family and multi-engine test cell are summarized in a table in order to better comprehension. In addition, the comparison between CFM56 family and multi-engine test cell has been performed in terms of time and cost.

Table 9-1: Conclusion

Engine Test Cell	Components	Multi-Engine Test Cell vs. Single Engine Test Cell
Civil Construction	Test Cell	For a multi-engine test cell the design of an indoor test cell should be performed considering the maximum mass flow and engine dimensions.
Test Cell Components	Acoustic System	For a multi-engine test cell the acoustic absorbents should be designed considering the maximum range of mass flow and consequently high amount of noise.
	Intake System	For a multi-engine test cell the intake system should be designed in order to reach the goal of testing different engine types.
	Exhaust system	For a multi-engine test cell the exhaust system should be designed in order to reach the goal of testing different engine types.
	Superstructure	For a multi-engine test cell the superstructure should be designed considering enough strength for withstanding the engine and its equipment with maximum weight.
	Engine Adapter	The adapter design should be performed based on engine type and thrust level. So, separated adapter will be dedicated to each different engine.
	Thrust Measurement System	For a multi-engine test cell the thrust stand and included load cells should be modified considering enough strength for withstanding the engine and its equipment with maximum weight and maximum thrust.
	Engine handling and Transportation	For a multi-engine test cell the transportation system should be designed with the maximum capability of engine handling.
	Lifting Platform	For a multi-engine test cell the lifting system should be designed with the maximum capability of engine handling.
	Vibration Damping system	For a multi-engine test cell the vibration damping system which is adapted with changing the engine type and to isolate the maximum level of vibration with sufficient effectiveness.
	Augmenter and Blast Basket	For a multi-engine test cell the augmenter tube and blast basket should be designed in order to reach the goal of testing different engine types.

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	Bellmouth	Specific bell-mouth must be provided for each engine with different dimensions.
Auxiliary Systems	Fuel System	Separated/one system must be provided for each engine with different characteristics such as type of fuel, required fuel capacity (pumps, heat exchangers, filters, piping, and valves), control systems and access points.
	Oil Filling System	Separated/one system or must be provided for each engine with different characteristics such as type of oil (to resistance to high temperature, prevent corrosion and changes in solution chemistry), required oil capacity (pumps, heat exchangers, filters, piping, valves), control systems and access points.
	Supply System Capabilities	This item included power supply, water supply, air supply etc. will be increased with increasing in thrust value and multi-engine test cell construction.
DAQ & Control System	Control System	Some hardware components are specific for each engine types and two specified software will be needed.
	Accuracy of Measurement	The thrust correlations should be calculated during separated measurement procedure for each engine.
Miscellany	Metrological Station	This system for a multi-engine test cell is the same as the system in a test cell for one engine family testing.
	Fire Extinguishing System	The capacity of fire distinguish system should be recalculate.
	CCTV & Intercom	This system for a multi-engine test cell is the same as the system in a test cell for one engine family testing.